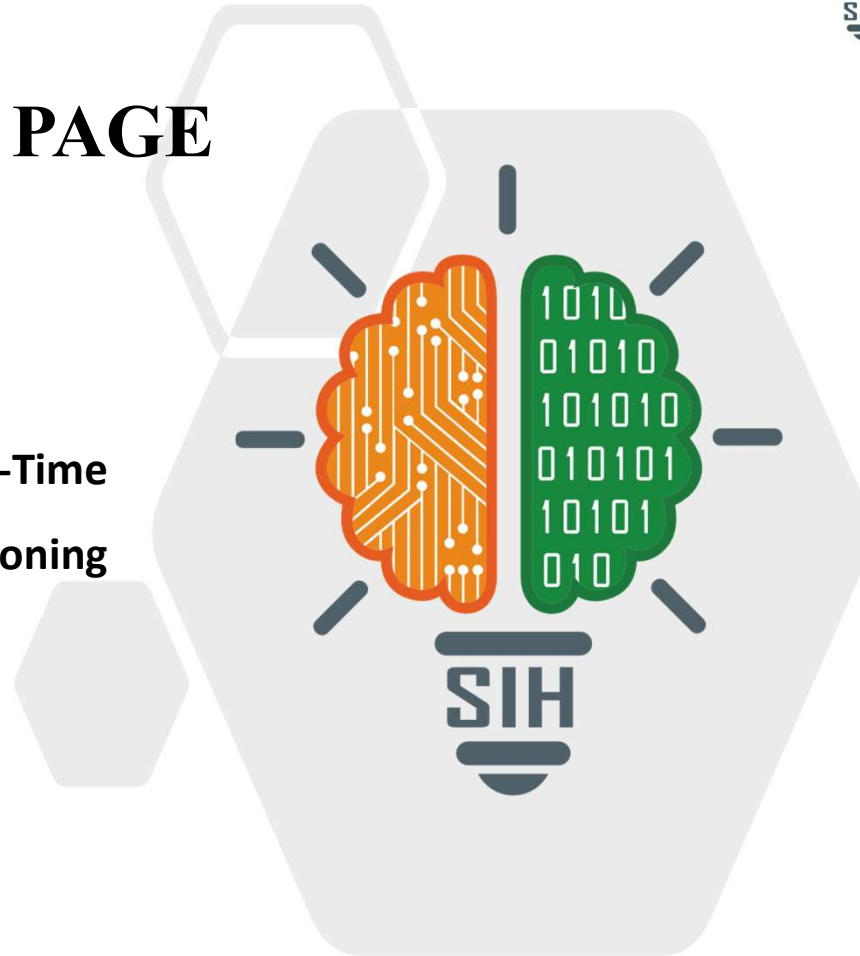


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID – SIH20614**
- **Problem Statement Title- AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks**
- **Theme- Blockchain & Cybersecurity**
- **PS Category- Software**
- **Team ID-**
- **Team Name: Code Nova**



Idea / Proposed Solution



The Problem

- AI voice-cloning tools can now replicate a person's voice from just 3–5 seconds of audio, freely available online.
- Fraudsters use cloned voices of relatives, executives, or officials to run vishing, extortion, and “emergency money” scams over phone calls.
- Victims cannot distinguish a real call from a synthetic one, causing rising financial and emotional harm across India.
- Existing spam/caller-ID systems check the number, not the voice — they cannot detect that the speaker itself is fake.



Our Solution — **VOVERA**

Real-time deepfake detection

Analyzes live audio streams using HuBERT transformer models to catch spectral anomalies unique to AI-generated speech.

Generative AI Security Copilot

Translates complex risk metrics into simple, human-readable alerts (via FLAN-T5) so users easily understand the exact threat.

Personal voiceprint matching

Validates incoming callers against locally encrypted, mathematical voice embeddings (ECAPA-TDNN) to stop impersonators.

Blockchain Audit Ledger

Cryptographically hashes and secures flagged call data, ensuring a mathematically tamper-proof evidence trail for compliance.

Technical Approach

System Workflow



Technology Stack

Core AI Engine: Built using PyTorch. Employs a HuBERT model (facebook/hubert-base-ls960) with a custom MLP classification head.

Speaker Verification: Custom ECAPA-TDNN architecture using 1D Convolutions to extract robust 192-D acoustic fingerprints.

Generative AI Explainer: Local FLAN-T5 transformer model to dynamically generate human-readable security alerts.

Modular Codebase: Organized into backend, frontend, ai, genai, and blockchain modules for clean separation of concerns.

Backend For Architecture: High-performance asynchronous FastAPI server utilizing WebSockets real-time continuous audio chunk streaming.

Frontend Dashboard: Developed using React 18, providing Dashboard, Live Call, and Investigations views with real-time risk meters and Security Copilot chat.

Data Security & Audit: SQLite/PostgreSQL with Fernet encryption. Custom blockchain ledger (SHA-256 hashing) seals analysis results for proof of fraud.

Development Setup: Python 3.10+ backend on SQLite (vigil_voice.db), launched via uvicorn with hot-reload for rapid iteration.

Feasibility and Viability

✓ Strengths

- **Ultra-Lightweight & Cost-Effective:** Entire AI ensemble uses low memory
- **Zero Network Latency for AI:** PyTorch models run directly within the FastAPI memory space, enabling true real-time analysis.
- **Comprehensive Public Datasets:** Leverages open-source datasets like ASVspoof and WaveFake for rapid model training.

⚠ Challenges

- **Voice-cloning tools evolve fast** — models must generalize to unseen methods
- **Real-time detection needs very low latency** without draining battery/data
- **Storing voiceprints raises privacy and consent concerns**

🛡 Mitigation Strategy

- **Continuous Retraining Pipeline:** Designed to ingest new synthetic audio to fine-tune the HuBERT classifier against zero-day threats.
- **Asynchronous Data Handling:** FastAPI utilizes asyncio to decouple WebSocket ingestion from heavy ML inference.
- **Encrypted Local Vector Storage:** Voiceprints are instantly converted into 192-D embeddings and secured with Fernet encryption.
- **Phased rollout:** banks/helplines first, then general consumer app



Impact and Benefits

3 sec

of audio is enough for current AI tools to clone a voice

24/7

real-time protection on every incoming call

Zero

extra hardware — works as a lightweight app/telecom layer



Protects Vulnerable Citizens

Shields vulnerable users from scams using plain-English GenAI warnings. Converts complex AI metrics into simple visual alerts.



Reduces Financial Fraud Losses

Automated first line of defense. Early detection of cloned-voice vishing prevents large-scale unauthorized wire transfers and UPI fraud.



Strengthens Telecom Trust

Gives telecom operators and financial institutions a verifiable security layer to guarantee communications are genuinely from the claimed official.



Supports Law Enforcement

Flagged calls log acoustic evidence into a Blockchain Audit Ledger, creating a mathematically proven chain of custody for prosecution.

Research and References

- 1 ASVspoof Challenge — benchmark datasets and baselines for audio anti-spoofing / deepfake detection research.
- 2 AASIST / RawNet2 — end-to-end deep-learning architectures for spoofed-speech detection used as model references.
- 3 CERT-In and TRAI public advisories on voice-cloning-enabled phone fraud and citizen safety guidelines.
- 4 RBI guidelines on digital-payment fraud prevention, relevant to voice-cloning-triggered banking scams.
- 5 Peer-reviewed literature on speaker verification embeddings (e.g., ECAPA-TDNN) for voiceprint matching.
- 6 News and industry reports documenting real-world AI voice-cloning fraud cases used to scope the problem statement.